Threshold policies for the fluid forgetting model

Analytical research note

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This note proves the threshold structure for the fluid model with the exact Beta demand. The argument uses no numerical approximation. It concerns the continuous-time control problem; a corresponding theorem for the original stochastic Bellman equation does not follow from this proof.

Derivation of the fluid model

Assume

$$0 < p_1 < p_2 < 1, \quad 0 \le c_1 < c_2 < R, \quad 0 < p_0 < 1, \quad 0 < \rho, \gamma < 1.$$

As in the fluid progress note, we use the normalized coordinate $x = (1-\rho)S \in [0,1]$ and write

$$\lambda = -\log \rho, \qquad \delta = -\log \gamma, \qquad \Delta p = p_2 - p_1, \qquad \Delta c = c_2 - c_1.$$

On the mature boundary, the probability of a purchase is

$$D(x) = 1 - I_{p_0}\left(1 + \frac{x}{1-\rho}, 1 + \frac{1-x}{1-\rho}\right).$$

Here $I_z(a,b)$ is the regularized incomplete beta function. Under product i, a purchase yields profit $R - c_i$ and succeeds with probability p_i. If no purchase occurs, the state stays at x.

Fluid Bellman for V

Write V for the value when the product is chosen optimally. Following the fluid progress note, consider the transitions over a small time step dt:

$$T^s_{dt}(x) = \rho^{dt} x + (1 - \rho^{dt}), \qquad T^f_{dt}(x) = \rho^{dt} x.$$

Over this step, the profit per purchase under product i is $(R - c_i)dt$ and the discount factor is γ^{dt}. We keep $D(x)$ fixed at the original ρ as $dt \to 0$. Since

$$\rho^{dt} = e^{-\lambda dt} = 1 - \lambda dt + o(dt),$$
$$\gamma^{dt} = e^{-\delta dt} = 1 - \delta dt + o(dt),$$

the state increments are

$$T^s_{dt}(x) - x = \lambda(1-x)dt + o(dt),$$
$$T^f_{dt}(x) - x = -\lambda x dt + o(dt).$$

Under product i, the expected change in the state is

$$D(x)\big[p_i\lambda(1-x)+(1-p_i)(-\lambda x)\big]dt+o(dt)=D(x)\lambda(p_i-x)dt+o(dt).$$

Thus the fluid state equation and profit rate are

$$\dot{x}=D(x)\lambda(p_i-x),\qquad D(x)(R-c_i). \tag{1}$$

To keep the notation simple, we use V at both the discrete and fluid stages of the following formal first-order derivation. The Bellman equation over dt is

$$\begin{aligned}V(x)=&(1-D(x))\gamma^{dt}V(x)\\&+D(x)\max_{i\in\{1,2\}}\left\{(R-c_i)dt+\gamma^{dt}\left[p_iV(T^s_{dt}(x))+(1-p_i)V(T^f_{dt}(x))\right]\right\}.\end{aligned} \tag{2}$$

Taylor expansion gives

$$\begin{aligned}V(T^s_{dt}(x))&=V(x)+\lambda(1-x)V'(x)dt+o(dt),\\V(T^f_{dt}(x))&=V(x)-\lambda xV'(x)dt+o(dt).\end{aligned}$$

Combining the success and failure terms,

$$p_iV(T^s_{dt}(x))+(1-p_i)V(T^f_{dt}(x))=V(x)+\lambda(p_i-x)V'(x)dt+o(dt).$$

Substituting this and $\gamma^{dt}=1-\delta dt+o(dt)$ into (2) and collecting terms,

$$V(x)=V(x)+\Big[D(x)\max_{i\in\{1,2\}}\{R-c_i+\lambda(p_i-x)V'(x)\}-\delta V(x)\Big]dt+o(dt).$$

Cancelling $V(x)$, dividing by dt, and letting $dt\to 0$ yields

$$\delta V(x)=D(x)\max_{i\in\{1,2\}}\{R-c_i+\lambda(p_i-x)V'(x)\}. \tag{3}$$

To obtain the discounted profit representation, let $i(t)$ denote any admissible product choice and let $x(t)$ follow (1), starting at $x(0)=x$. Since (3) takes the maximum over products, wherever V is differentiable we have

$$\delta V(x(t))\geq D(x(t))(R-c_{i(t)})+V'(x(t))\dot{x}(t).$$

Multiplying by $e^{-\delta t}$ and using the chain rule gives

$$\begin{aligned}\frac{d}{dt}\big[e^{-\delta t}V(x(t))\big]&=e^{-\delta t}\big[V'(x(t))\dot{x}(t)-\delta V(x(t))\big]\\&\leq -e^{-\delta t}D(x(t))(R-c_{i(t)}).\end{aligned}$$

Integrating from 0 to T,

$$e^{-\delta T}V(x(T))-V(x)\leq-\int_0^T e^{-\delta t}D(x(t))(R-c_{i(t)})\,dt.$$

Rearranging,

$$V(x)\geq\int_0^T e^{-\delta t}D(x(t))(R-c_{i(t)})\,dt+e^{-\delta T}V(x(T)).$$

Because V is bounded and $\delta > 0$, the terminal term tends to zero as $T \to \infty$. Thus $V(x)$ is at least the discounted profit of every admissible policy. For an admissible policy attaining the Bellman maximum along its trajectory, these inequalities become equalities. This includes holding mixtures of tied maximizing products, as described below. The chain-rule calculation applies where V is differentiable; at corners, dynamic programming supplies the corresponding argument. Consequently, the fluid value is the largest discounted profit over these trajectories:

$$V(x) = \sup \int_0^\infty e^{-\delta t} D(x(t))(R - c_{i(t)})\, dt, \qquad x(0) = x. \tag{4}$$

We also allow mixtures: both the drift and the profit rate are averaged using the same product fractions. In particular, at $x \in [p_1, p_2]$, the product-2 fraction

$$\frac{x - p_1}{p_2 - p_1}$$

makes the average drift zero and holds the state at x. Maximizing over mixtures gives the same Bellman equation, because both drift and profit depend linearly on the fractions.

Where V is differentiable, product 2 is strictly preferred when

$$\begin{aligned} R - c_2 + \lambda(p_2 - x)V'(x) &> R - c_1 + \lambda(p_1 - x)V'(x) \\ \iff \qquad V'(x) &> \frac{\Delta c}{\lambda \Delta p}. \end{aligned}$$

The Taylor derivation applies where the derivatives exist; the threshold proof below does not assume that V is differentiable everywhere. The proposition establishes the global policy structure for this fluid problem, including holding mixtures at attractive thresholds.

Proposition 1 (At most one interval of product 2)**.** *The fluid problem admits an optimal stationary policy with numbers $0 \le x_L \le x_H \le 1$ such that, away from threshold states,*

$$i(x) = \begin{cases} 1, & x < x_L, \\ 2, & x_L < x < x_H, \\ 1, & x > x_H. \end{cases}$$

Empty intervals are allowed. Thus the nonconstant action patterns, in increasing order of the state, are

$$1 \longrightarrow 2, \qquad 2 \longrightarrow 1, \qquad 1 \longrightarrow 2 \longrightarrow 1.$$

Constant policies are also possible. There cannot be two separated regions of strict preference for product 2, such as a strict $2 \to 1 \to 2$ pattern. At an attractive threshold in (p_1, p_2) the optimal prescription includes the holding mixture. Choices at indifference points are made consistently with the stated interval policy.

The proof has two parts. First, a shape property of the Beta demand limits the transformed payoff to at most one valley followed by one peak. Second, the one-dimensional control problem then has at most one interval on which product 2 is optimal. Neither part assumes that the optimal value is continuously differentiable everywhere.

1. An exact transformation

We want to rewrite the objective so that the product choice enters only through the movement of the state. To find the right adjustment, compare the two products at the same state x. Choosing product 2 reduces the current profit rate by $D(x)\Delta c$ and increases the state drift by $\lambda D(x)\Delta p$. Their ratio is

$$\frac{D(x)\Delta c}{\lambda D(x)\Delta p} = \frac{\Delta c}{\lambda \Delta p}.$$

This ratio is independent of x. It suggests crediting each extra unit of state accumulation at the rate $\Delta c/(\lambda \Delta p)$, so that the extra drift exactly offsets the extra product cost.

To see the cancellation explicitly, the two products satisfy

$$c_i = c_1 + \frac{\Delta c}{\Delta p}(p_i - p_1), \qquad i \in \{1, 2\}.$$

Adding the state credit to the profit rate and using $\dot{x} = \lambda D(x)(p_i - x)$ gives

$$\begin{aligned} D(x)(R - c_i) + \frac{\Delta c}{\lambda \Delta p}\dot{x} &= D(x)\left[R - c_i + \frac{\Delta c}{\Delta p}(p_i - x)\right] \\ &= D(x)\left[R - c_1 + \frac{\Delta c}{\Delta p}(p_1 - x)\right]. \end{aligned}$$

The last expression is the same for either product. Rearranging,

$$D(x)(R - c_i) = D(x)\left[R - c_1 + \frac{\Delta c}{\Delta p}(p_1 - x)\right] - \frac{\Delta c}{\lambda \Delta p}\dot{x}.$$

The first term now depends only on the state; the product choice affects the second term through $\dot{x}$. The same identity holds for mixtures, since both the drift and the profit are averaged with the same fractions.

Next, remove the term involving $\dot{x}$ from the discounted objective. For a trajectory starting at $x(0) = x$, integration by parts gives

$$\begin{aligned} \int_0^T e^{-\delta t}\dot{x}(t)\,dt &= \left[e^{-\delta t}x(t)\right]_0^T + \delta \int_0^T e^{-\delta t}x(t)\,dt \\ &= e^{-\delta T}x(T) - x + \delta \int_0^T e^{-\delta t}x(t)\,dt. \end{aligned}$$

Since $x(T) \in [0, 1]$ and $\delta > 0$, the terminal term vanishes as $T \to \infty$. Therefore,

$$-\frac{\Delta c}{\lambda \Delta p}\int_0^\infty e^{-\delta t}\dot{x}(t)\,dt = \frac{\Delta c}{\lambda \Delta p}x - \frac{\delta \Delta c}{\lambda \Delta p}\int_0^\infty e^{-\delta t}x(t)\,dt.$$

Combining this with the rewritten profit rate identifies the payoff that remains inside the integral:

$$F(x) = D(x)\left[R - c_1 + \frac{\Delta c}{\Delta p}(p_1 - x)\right] - \frac{\delta \Delta c}{\lambda \Delta p}x. \tag{5}$$

Thus the first term in F is the common payoff obtained after crediting state accumulation. The second is the discount correction generated by integration by parts. This is how the definition of F is obtained.

For every admissible policy, its discounted profit equals the initial state credit plus the discounted integral of F along its trajectory. Taking the supremum over policies yields

$$V(x) = \frac{\Delta c}{\lambda \Delta p} x + \sup \int_0^\infty e^{-\delta t} F(x(t))\, dt. \tag{6}$$

For a given initial state, the first term is the same for every policy, so the transformation preserves their ranking.

For the proof, write $W(x) = V(x) - \Delta c\, x/(\lambda \Delta p)$. Then W is the value of maximizing the discounted integral of F. Where the derivative exists,

$$V'(x) = W'(x) + \frac{\Delta c}{\lambda \Delta p}.$$

Substituting into (3) and using the same cancellation gives its control equation:

$$\delta W(x) = F(x) + \max_{i=1,2} \{\lambda D(x)(p_i - x) W'(x)\}. \tag{7}$$

Where the derivative exists, product 2 is strictly optimal precisely when $W' > 0$, and product 1 is strictly optimal when $W' < 0$. The transformed flow payoff is the same for both products at a given state. Products determine the direction and speed of movement through the graph of F, which is why studying the shape of this single function helps us characterize the optimal policy.

2. The Beta demand has the required shape

Lemma 1 (Shape of the exact Beta demand)**.** *Let $0 < p_0 < 1$, $0 < \rho < 1$, and $n = (1-\rho)^{-1}$. For*

$$D(x) = 1 - I_{p_0}\big(1 + nx, 1 + n(1-x)\big), \qquad 0 \le x \le 1,$$

we have $D'(x) > 0$ and

$$D'(x) D'''(x) - [D''(x)]^2 < 0.$$

In particular, D' is strictly log-concave. The endpoint derivatives are understood through the smooth extension of D.

Proof. We establish a fixed-total Beta identity first. In this proof only, fix $N > 2$ and $p \in (0,1)$, write $b = N - a$, and let

$$S(a) = 1 - I_p(a, b), \qquad 0 < a < N.$$

For $0 < h < b$, take a Dirichlet random vector (U, Y, W) with parameters $(a, h, b-h)$. Then U has the Beta(a,b) distribution, whereas $U + Y$ has the Beta$(a+h, b-h)$ distribution. Since $U + Y = 1 - W \ge U$,

$$S(a+h) - S(a) = \Pr(U < p,\ W < 1 - p).$$

Integrating the Dirichlet density over this rectangle, dividing by h, and letting $h \downarrow 0$ gives

$$S'(a) = \frac{1}{B(a,b)} \int_0^p \int_0^{1-p} \frac{u^{a-1} w^{b-1}}{1-u-w}\, dw\, du > 0. \tag{8}$$

Here $h\Gamma(h) \to 1$. Dominated convergence is valid: the corner singularity is proportional to $\big[(p-u) + (1-p-w)\big]^{-1}$, which is integrable in two dimensions. Choosing a fixed $h_0 < b$ bounds the remaining integrand by a constant times $u^{a-1} w^{b-h_0-1}/(1-u-w)$, also integrable.

Expand $(1-u-w)^{-1}$ into its nonnegative geometric series and use the binomial theorem. Tonelli's theorem applied to (8) yields

$$S'(a) = \sum_{j,k\geq 0} t_{jk}(a), \qquad t_{jk}(a) = \frac{p^a(1-p)^b}{B(a,b)}\binom{j+k}{j}\frac{p^j(1-p)^k}{(a+j)(b+k)}. \tag{9}$$

Let expectations and variances below use the probability weights $t_{jk}(a)/S'(a)$. With ψ and ψ_1 denoting the digamma and trigamma functions, respectively, differentiation at fixed $a+b=N$ gives

$$(\log t_{jk})' = \log\frac{p}{1-p} - \psi(a) + \psi(b) - \frac{1}{a+j} + \frac{1}{b+k},$$

and

$$(\log t_{jk})'' = -\psi_1(a) - \psi_1(b) + \frac{1}{(a+j)^2} + \frac{1}{(b+k)^2}.$$

The logarithmic derivative of a positive sum therefore satisfies

$$\begin{aligned}(\log S')'' = & -\psi_1(a) - \psi_1(b) + \mathbb{E}\left[\frac{1}{(a+j)^2} + \frac{1}{(b+k)^2}\right] \\ & + \operatorname{Var}\left(-\frac{1}{a+j} + \frac{1}{b+k}\right).\end{aligned}$$

Termwise differentiation is justified on compact subintervals of $(0,N)$: the summands and their first two derivatives are bounded by a constant times the convergent positive summands at any fixed interior parameter.

The variable in the variance lies in $[-1/a, 1/b]$. Using the elementary bound $\operatorname{Var}(Z) \leq (M-m)^2/4$ when $Z \in [m,M]$, we obtain

$$(\log S')'' \leq -\psi_1(a) - \psi_1(b) + \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{4}\left(\frac{1}{a} + \frac{1}{b}\right)^2. \tag{10}$$

For every $s > 0$, the trigamma series satisfies

$$\psi_1(s) = \sum_{m=0}^{\infty} \frac{1}{(s+m)^2} > \frac{1}{s} + \frac{1}{2s^2}.$$

Indeed, sum the strict trapezoidal inequality for the strictly convex function $t \mapsto (s+t)^{-2}$ over $[m, m+1]$, $m \geq 0$.

Suppose now that $a, b \geq 1$. Put $A = 1/a$ and $B = 1/b$ for the following algebra, so that $0 < A, B \leq 1$. Equation (10) gives

$$\begin{aligned}(\log S')'' &< -(A+B) + \frac{3}{4}(A^2+B^2) + \frac{1}{2}AB \\ &\leq -\frac{1}{4}(A^2+B^2) + \frac{1}{2}AB = -\frac{1}{4}(A-B)^2 \leq 0,\end{aligned}$$

where the second inequality uses $A + B \geq A^2 + B^2$. The first inequality is strict, including when $A = B$.

Finally, take $N = n+2$, $p = p_0$, and $a = 1 + nx$. Both a and $b = 1 + n(1-x)$ are at least one on $[0,1]$. Since $D'(x) = nS'(1+nx)$, positivity and strict log-concavity follow from the preceding calculation. □

Lemma 2 (Shape of the transformed payoff)**.** *The set* $\{x \in [0,1] : F'(x) > 0\}$ *is an interval, possibly empty or meeting a boundary. If nonempty,* F *decreases before this interval, increases within it, and decreases after it, with absent segments omitted. If it is empty, product 1 is optimal everywhere.*

Proof. For this proof, put

$$q = p_1 + \frac{(R - c_1)\Delta p}{\Delta c} > p_2.$$

Differentiating (5) gives

$$F'(x) = \frac{\Delta c}{\Delta p}\left[(q - x)D'(x) - D(x) - \frac{\delta}{\lambda}\right].$$

Since $D > 0$ and $D' > 0$, we have $F'(x) < 0$ for $x \geq q$.

Let $I = \{x \in [0,1] : F'(x) > 0\}$. Suppose, for a contradiction, that I is not an interval. Then there exist $x_1 < x_2 < x_3$ in $[0,1]$ such that

$$F'(x_1) > 0, \qquad F'(x_2) \leq 0, \qquad F'(x_3) > 0.$$

The preceding observation implies $x_3 < q$. By the mean value theorem, there exist $u \in (x_1, x_2)$ and $v \in (x_2, x_3)$ such that

$$F''(u) = \frac{F'(x_2) - F'(x_1)}{x_2 - x_1} < 0, \qquad F''(v) = \frac{F'(x_3) - F'(x_2)}{x_3 - x_2} > 0.$$

On the other hand, define

$$r(x) = \frac{D''(x)}{D'(x)}.$$

Lemma 1 shows that r is strictly decreasing, and a second differentiation gives

$$F''(x) = \frac{\Delta c}{\Delta p} D'(x)\big[r(x)(q - x) - 2\big]. \tag{11}$$

Since $F''(v) > 0$ and $v < q$, we obtain

$$r(v)(q - v) > 2, \qquad r(v) > 0.$$

But $u < v < q$, so $r(u) > r(v) > 0$ and $q - u > q - v > 0$. Consequently,

$$r(u)(q - u) > r(v)(q - v) > 2.$$

Equation (11) now gives $F''(u) > 0$, contradicting $F''(u) < 0$. Thus I is an interval, possibly empty or meeting a boundary.

We have shown that $I = \{x : F'(x) > 0\}$ is an interval, possibly empty. Suppose now that I is nonempty. It remains to rule out additional zeros of F' outside I. Let z be a zero of F' to the left of I, and choose $y \in I$. By the mean value theorem, there exists $v \in (z, y)$ with $F''(v) > 0$. Repeating the comparison used above then gives $F''(z) > 0$. Hence F' crosses zero from negative to positive at z, so z must be the left boundary of I. Similarly, every zero to the right of I satisfies $F''(z) < 0$ and must be its right boundary. Therefore $F' < 0$ outside the closure of I, and F decreases before I, increases on I, and decreases afterward.

If I is empty, F is nonincreasing. The smallest admissible trajectory, produced by product 1, therefore maximizes F at every time, so product 1 is optimal everywhere. □

3. Why this shape excludes additional policy intervals

Lemma 3 (A single interval of use of product 2)**.** *Suppose $0 < p_1 < p_2 < 1$, and $\delta, \lambda > 0$. Allow relaxed controls, so that any state in $[p_1, p_2]$ can be held fixed. Suppose that, for some $0 \le a < m \le 1$,*

$$F'(x) < 0 \quad \text{on } (0, a) \cup (m, 1), \qquad F'(x) > 0 \quad \text{on } (a, m), \tag{12}$$

with any interior zeros of F' being simple. Then an optimal policy can be selected which uses product 2 on at most one interval and product 1 outside it. Thus, as the state increases, the only nonconstant action patterns are

$$1 \to 2, \qquad 2 \to 1, \qquad 1 \to 2 \to 1. \tag{13}$$

At an attractive threshold in (p_1, p_2), the optimal policy may use the mixture that holds the state fixed.

The monotone cases are immediate. If F is nonincreasing, product 1 is optimal everywhere. If F is nondecreasing, product 2 is optimal everywhere.

Proof. Throughout this proof write $b_i(x) = \lambda D(x)(p_i - x)$ and $g(x) = \delta W(x) - F(x)$. The transformed Bellman equation is

$$\delta W = F + \max\{b_1 W', b_2 W'\}.$$

The trajectory under product 1 is the smallest admissible trajectory at every time, and the trajectory under product 2 is the largest. This proves the two monotone cases directly. We now consider the stated decrease–increase– decrease shape.

The signs of the drifts $b_i(x)$ are determined by the position of x, specifically:

$$b_1(x) > 0\,, \quad b_2(x) > 0 \quad \text{for } x < p_1 \tag{14}$$

$$b_1(x) < 0\,, \quad b_2(x) > 0 \quad \text{for } p_1 < x < p_2 \tag{15}$$

$$b_1(x) < 0\,, \quad b_2(x) < 0 \quad \text{for } x > p_2 \tag{16}$$

Hence, for any branch of product i we have that the sign of g is related to the sign of W', specifically:

$$\operatorname{sign}(g) = \operatorname{sign}(W')\,, \qquad \text{for } x < p_1 \tag{17}$$

$$\operatorname{sign}(g) = -\operatorname{sign}(W')\,, \quad \text{for } x > p_2 \tag{18}$$

Crossings outside $[p_1, p_2]$. Both drifts are positive below p_1 and negative above p_2. Recall that

$$g(x) = \max_{i \in \{1,2\}} \{b_i(x) W'(x)\}$$

On a branch where product i is used, we have $g(x) = b_i(x) W'(x)$, hence

$$W'(x) = \frac{g(x)}{b_i(x)} = \frac{\delta W(x) - F(x)}{b_i(x)}.$$

Since F and b_i are smooth and b_i is locally bounded away from zero, the right-hand side is locally Lipschitz in (x, W). Moreover, the two branches coincide when $g = 0$. Hence the associated one-dimensional ODE has a unique local solution, and W is continuously differentiable outside $[p_1, p_2]$.

On a branch, we have

$$g' = \frac{\delta}{b_i} g - F'$$

At an interior zero of g with $F' \neq 0$, the sign of g' is the sign of $-F'$. Since the signs of g and W' agree below p_1 and are opposite above p_2, the possible strict action crossings are

	$F' < 0$	$F' > 0$
$x < p_1$	$1 \to 2$	$2 \to 1$
$x > p_2$	$2 \to 1$	$1 \to 2.$

(*)

If $g(x^*) = 0$ at a point where $F'(x^*) = 0$, with $x^* \notin \{p_1, p_2\}$, then

$$g'(x^*) = 0.$$

Differentiating the branch equation and using $g(x^*) = g'(x^*) = 0$ gives

$$g''(x^*) = -F''(x^*) \neq 0,$$

where the last inequality follows because the interior zeros of F' are simple. Hence g touches zero without changing sign, so the strict product preference does not change at x^*. The behavior at p_1 and p_2 requires a separate endpoint argument, since one of the drifts vanishes there. This part is covered in the end of the proof.

Comparison of destinations inside $[p_1, p_2]$. Suppose first that $m > p_1$, and put $h = \min\{m, p_2\}$ for this proof.

Let $L(x)$ be the transformed value of always using product 1, approaching p_1. Let $H(x)$ be the transformed value of using product 2 to reach h and then holding there. When $h = p_2$, this means approaching p_2 asymptotically under product 2. On (p_1, h),

$$\delta L = F + b_1 L', \qquad \delta H = F + b_2 H', \qquad L(p_1) = \frac{F(p_1)}{\delta}, \quad H(h) = \frac{F(h)}{\delta}.$$

Let $x_i(t; x)$ denote the trajectory starting from x under product i,so that

$$\dot{x}_i(t; x) = b_i(x_i(t; x)), \qquad x_i(0; x) = x.$$

The preceding ODEs admit the corresponding discounted-payoff representations

$$L(x) = \int_0^\infty e^{-\delta t} F(x_1(t; x))\, dt.$$

If $h < p_2$, let $\tau_h(x)$ denote the time at which the product-2 trajectory reaches h. Then

$$H(x) = \int_0^{\tau_h(x)} e^{-\delta t} F(x_2(t; x))\, dt + e^{-\delta \tau_h(x)} \frac{F(h)}{\delta}.$$

When $h = p_2$, the product-2 trajectory approaches p_2 asymptotically, and the representation becomes

$$H(x) = \int_0^\infty e^{-\delta t} F(x_2(t; x))\, dt.$$

If x is in the decreasing part of F on this interval, the trajectory toward p_1 satisfies

$$F(x_1(t; x)) > F(x), \qquad t > 0,$$

and therefore

$$L(x) > \int_0^\infty e^{-\delta t} F(x)\, dt = \frac{F(x)}{\delta}.$$

Hence $\delta L(x) > F(x)$. If x is in the increasing part, the analogous comparison along the trajectory toward h gives $\delta H(x) > F(x)$. At the valley either comparison is strict. Thus

$$\delta \max\{L(x), H(x)\} > F(x), \qquad p_1 < x < h.$$

Where $L(x) > H(x)$, the preceding inequality implies $\delta L(x) - F(x) > 0$. Since $b_1(x) < 0$ on (p_1, h),

$$L'(x) = \frac{\delta L(x) - F(x)}{b_1(x)} < 0.$$

With this, and since $b_2(X) > 0$ we have $b_1 L' > b_2 L'$. Hence, product 1 maximizes the Bellman expression. Similarly where $H(x) > L(x)$, since $b_2(x) > 0$ we have

$$H'(x) = \frac{\delta H(x) - F(x)}{b_2(x)} > 0,$$

Recalling that $b_1(x) < 0$ we have $b_2 H(x) > b_1 H(x)$. Hence, product 2 is maximizes the bellman expression.

Now consider an interior crossing $x^* \in (p_1, h)$ such that $L(x^*) = H(x^*)$. Using the two ODEs,

$$(H - L)'(x^*) = \big(\delta L(x^*) - F(x^*)\big) \left(\frac{1}{b_2(x^*)} - \frac{1}{b_1(x^*)} \right) > 0.$$

Hence every interior crossing has the same orientation: L wins immediately to the left and H immediately to the right. There can therefore be at most one such crossing. If it occurs at x^*, then

$$L(x) > H(x), \qquad p_1 < x < x^*,$$

and

$$H(x) > L(x), \qquad x^* < x < h.$$

Moreover, following either winning branch remains in its winning region: under product 1 the state moves toward p_1, while under product 2 it moves toward h. Endpoint ties at p_1 and p_2 are treated separately below.

We now verify that $\widehat{W}$ is the optimal value in $[p_1, h]$. Define

$$\widehat{W}(x) = \max\{L(x), H(x)\}, \qquad x \in [p_1, h].$$

We first show that, for any initial state in $[p_1, h]$, no value is lost by restricting attention to trajectories that remain in this interval. If $h = p_2$, this follows immediately because $[p_1, p_2]$ is invariant. Suppose instead that $h = m < p_2$. Any excursion above h can be replaced by holding the state at h until the original trajectory returns. Such holding is feasible because $h \in (p_1, p_2)$ and relaxed controls are allowed. Since $h = m$ is the peak of F and F decreases for $x > h$,

$$F(x) \le F(h), \qquad x > h.$$

Therefore, this replacement weakly increases the discounted payoff. If the trajectory never returns to h, holding at h forever is likewise weakly better. Hence no optimal trajectory starting in $[p_1, h]$ needs to leave this interval.

We may therefore restrict the problem to $[p_1, h]$. By the preceding comparison, $\widehat{W}$ is continuous and continuously differentiable except possibly at the unique interior crossing of L and H. Moreover,

$$\delta \widehat{W}(x) \ge F(x), \qquad x \in [p_1, h], \tag{19}$$

with strict inequality for $p_1 < x < h$.

At every interior point where $L \neq H$, the winning branch satisfies its Bellman equation and selects the maximizing product. Hence

$$\delta\widehat{W}(x) = F(x) + \max_{i=1,2}\{b_i(x)\widehat{W}'(x)\}.$$

Since the drift under a mixture is a weighted average of $b_1(x)$ and $b_2(x)$, it follows that along any admissible trajectory in $[p_1, h]$,

$$\frac{d}{dt}\widehat{W}(x(t)) \leq \delta\widehat{W}(x(t)) - F(x(t))$$

for almost every t. If the trajectory spends positive time at an endpoint or at the possible crossing, the same inequality follows directly from (19).

Therefore,

$$\frac{d}{dt}\left[e^{-\delta t}\widehat{W}(x(t))\right] \leq -e^{-\delta t}F(x(t))$$

for almost every t. Integrating from 0 to T gives

$$\int_0^T e^{-\delta t}F(x(t))\,dt \leq \widehat{W}(x) - e^{-\delta T}\widehat{W}(x(T)).$$

Since $\widehat{W}$ is bounded on $[p_1, h]$, the terminal term vanishes as $T \to \infty$. Thus every admissible trajectory satisfies

$$\int_0^\infty e^{-\delta t}F(x(t))\,dt \leq \widehat{W}(x),$$

and therefore

$$W(x) \leq \widehat{W}(x).$$

Conversely, the policies defining L and H are admissible, so

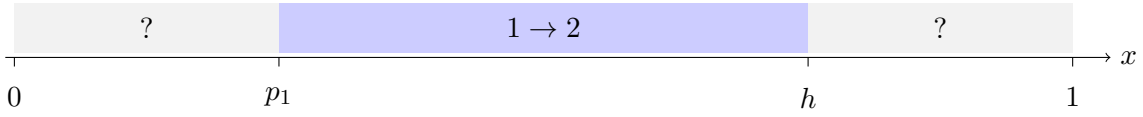
$$W(x) \geq L(x), \qquad W(x) \geq H(x).$$

Hence

$$W(x) = \widehat{W}(x) = \max\{L(x), H(x)\}, \qquad x \in [p_1, h].$$

Following the winning branch attains this value and, by the crossing orientation established above, remains in its winning region. Thus these choices define an optimal stationary policy on $[p_1, h]$.

The following figure illustrates what we have proved until now, that is,if is there any crossings in the interval $[p_1, h]$ is only in the direction $1 \to 2$. Constant policies are allowed too.



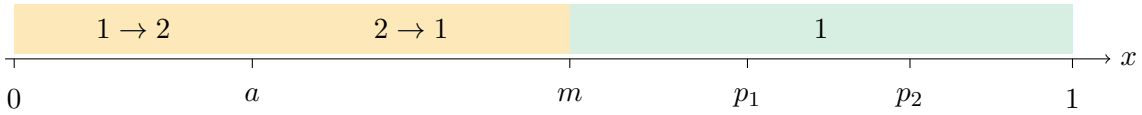
Case $m \leq p_1$. In this case we study the regions $x \geq m$ and $x < m$ separately.

For any $x \geq m$ every admissible trajectory stays in $[m, 1]$ given that $m \leq p_1 \leq p_2$ and so neither product can move the state across m from above. Since F decreases on this interval, product 1 is optimal there by trajectory ordering. For $x < m$, since $m \leq p_1$, this entire region lies below p_1. On $(0, a)$, where $F' < 0$, Table $(*)$ permits only a $1 \to 2$ crossing, which implies that there can only exist

one such crossing, given that after switching to product 2, returning to product 1 would require a $2 \to 1$ crossing, which is not possible on this interval. On (a, m), where $F' > 0$, the only permitted crossing is $2 \to 1$, so again there can be at most one.

Consequently, the only possible nonconstant pattern below m is $1 \to 2 \to 1$ with either crossing possibly absent. Since product 1 is optimal from m onward, product 2 is therefore optimal on at most one interval. The same argument applies when $m = p_1$, where product 1 can hold the state at the peak.

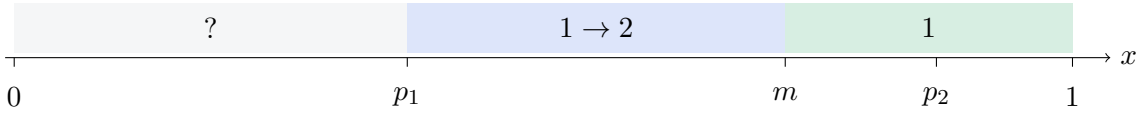
In this case we have prooved the only crossings that can happen are represented as follows:



Case $p_1 < m \leq p_2$. Above m, product 1 is optimal. Fix $x > m$ and compare any admissible trajectory with the trajectory that uses product 1 until reaching m. Since $b_1(x) < b_2(x)$ at every state, the product-1 trajectory is the smallest admissible trajectory and therefore reaches m no later than the comparison trajectory. While both trajectories remain above m, F is decreasing, so the product-1 trajectory yields a weakly higher flow payoff.

If the product-1 trajectory reaches m first, it can hold the state at m until the comparison trajectory arrives. During this time it earns $F(m)$, which is at least the flow payoff of any state above m. Once both trajectories reach m, it can follow the same continuation. If the comparison trajectory never reaches m, holding at m forever is likewise weakly better. Hence any policy starting from $x > m$ is weakly dominated by one that uses product 1 until reaching m and then continues optimally. Therefore, product 1 is optimal for every $x > m$.

On $[p_1, m]$, the preceding destination comparison gives at most one $1 \to 2$ crossing. We have proved then:



Now we prove that there cannot be a detached product-2 interval below p_1. Suppose first that $a > p_1$. Then $F' < 0$ below p_1, so table $(*)$ allows only $1 \to 2$ crossings in this region. At p_1, if the low branch wins we have

$$W(p_1) = L(p_1) = \frac{F(p_1)}{\delta},$$

So

$$g(p_1) = \delta W(p_1) - F(p_1) = 0$$

If product 2 were optimal immediately to the left of p_1 we would have $b_2 W' > b_1 W'$. Since $0 < b_1 < b_2$ in this region, that would imply $W' > 0$. Hence

$$g = b_2 W' > 0.$$

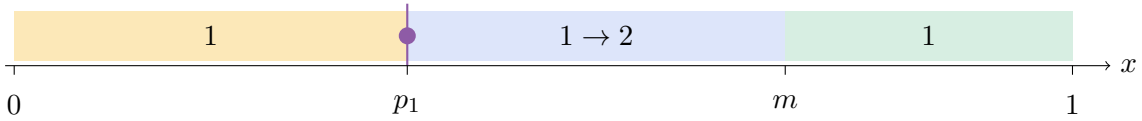
On this branch,

$$g' = \frac{\delta g}{b_2} - F' > 0,$$

because $g > 0$, $b_2 > 0$, and $F' < 0$ in this region. Hence $g' > 0$, so g is strictly increasing as x approaches p_1 from the left. This is incompatible with $g > 0$ immediately to the left of p_1 and

$g(p_1) = 0$. Therefore, product 2 cannot be optimal immediately to the left of p_1, and product 1 must be optimal there.

Since $a > p_1$, we have $F' < 0$ throughout $(0, p_1)$, so Table $(*)$ allows only $1 \to 2$ crossings in this region. We have just shown that product 1 is optimal immediately to the left of p_1. Therefore, product 2 cannot be optimal at any lower state, otherwise, the policy would have to switch from product 2 back to product 1 before reaching p_1, requiring a $2 \to 1$ crossing, which is impossible.



If instead the high branch strictly wins at p_1, then

$$W(p_1) = H(p_1) > L(p_1) = \frac{F(p_1)}{\delta},$$

so $g(p_1) = \delta W(p_1) - F(p_1) > 0$. By continuity, $g > 0$ immediately to the left of p_1 and since $0 < b_1(x) < b_2(x)$ we have that

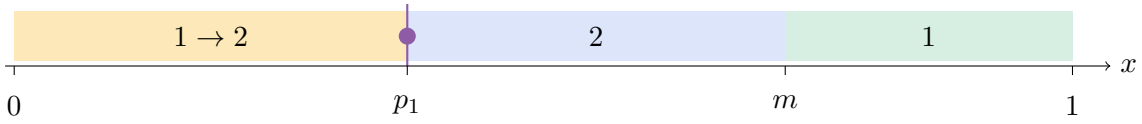
$$g(x) = \max\{b_1(x)W'(x),\, b_2(x)W'(x)\} > 0 \quad \Rightarrow \quad W'(x) > 0$$

Therefore,

$$b_2(x)W'(x) > b_1(x)W'(x),$$

so product 2 is strictly optimal immediately to the left of p_1.

Because $a > p_1$, Table $(*)$ allows only $1 \to 2$ crossings below p_1. Hence any product-2 region on the left can begin at most once and must continue up to p_1. Since the high branch also uses product 2 immediately to the right of p_1, the two regions are connected and therefore form a single product-2 interval. Endpoint ties are treated separately below.



Now suppose that $a < p_1$. Since $p_1 < m$, we have $F' > 0$ throughout (a, p_1). For any x in this interval, product 1 moves the state upward toward p_1, because $b_1(x) > 0$. Along this trajectory the state remains in (a, p_1), where F is strictly increasing. Hence the subsequent flow payoffs are strictly larger than $F(x)$, so

$$L(x) > \frac{F(x)}{\delta}.$$

Since by definition we have that $W(x) \geq L(x)$, it implies

$$g(x) = \delta W(x) - F(x) > 0.$$

Moreover, below p_1 we have $0 < b_1(x) < b_2(x)$, so we have

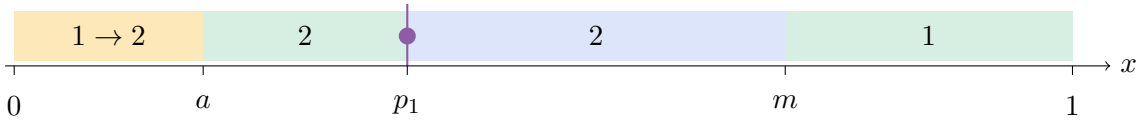
$$g(x) = \max\{b_1(x)W'(x),\, b_2(x)W'(x)\} > 0 \quad \Rightarrow \quad W'(x) > 0$$

Therefore,

$$b_2(x)W'(x) > b_1(x)W'(x),$$

So product 2 is strictly optimal throughout (a, p_1).

Finally, on $(0, a)$ we have $F' < 0$, so Table $(*)$ allows only $1 \to 2$ crossings. Since product 2 is already optimal immediately to the right of a, any product-2 region below a must connect to the one on (a, p_1) and it cannot form a separate interval.



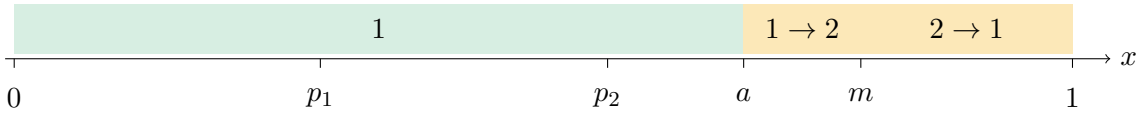
Finally, suppose that $a = p_1$, so that p_1 is the valley of F. Holding at p_1 yields the value $F(p_1)/\delta$, whereas moving toward the higher destination yields strictly larger subsequent values of F. Hence the high branch strictly wins at p_1, so $g(p_1) > 0$. By continuity, product 2 is also optimal immediately to the left of p_1. Since $F' < 0$ below p_1, Table $(*)$ allows only $1 \to 2$ crossings there, and any product-2 region on the left must therefore attach to the one at p_1. Hence product 2 can form at most one interval below and across p_1.

The conclusion of this part can be illustrated it as follows:



Case $m > p_2$. On $[p_1, p_2]$, the destination comparison applies with $h = p_2$, and the preceding argument below p_1 is unchanged.

Suppose first that $a \geq p_2$. Any trajectory starting at $x \leq a$ remains below a, where F is decreasing. Since product 1 generates the smallest admissible trajectory, it is optimal throughout this region. For $x > a$, table $(*)$ allows only $1 \to 2$ crossings on (a, m) and only $2 \to 1$ crossings on $(m, 1)$. Hence product 2 can be optimal on at most one interval. This case is illustrated as follows:



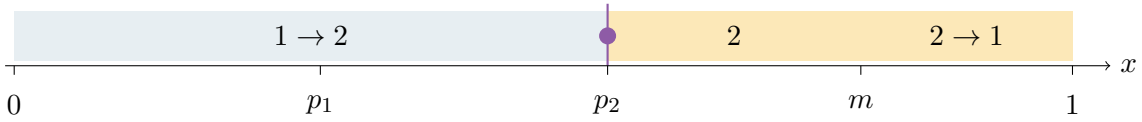
Now suppose that $a < p_2$. Then $F' > 0$ on (p_2, m), so table $(*)$ allows only $1 \to 2$ crossings there. If the high destination strictly wins at p_2, then

$$W(p_2) = H(p_2) = \frac{F(p_2)}{\delta},$$

and hence $g(p_2) = 0$. Product 1 immediately to the right would require $g > 0$. On that branch, however,

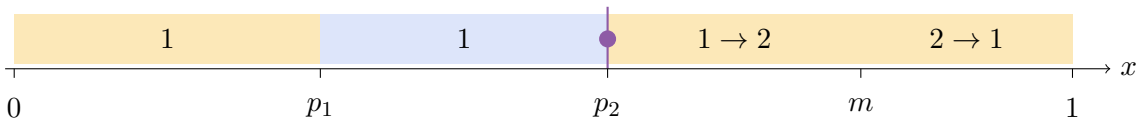
$$g' = \frac{\delta g}{b_1} - F' < 0,$$

since $b_1 < 0$ and $F' > 0$. Thus a positive g cannot emerge from $g(p_2) = 0$, so product 2 remains optimal immediately to the right of p_2. Since no $2 \to 1$ crossing is possible before m, this product-2 region cannot split before reaching m.



If instead the low destination strictly wins at p_2, then L must win throughout $[p_1, p_2]$. Indeed, any interior crossing between L and H has orientation $L \to H$. Hence, if H were to win at some state to the left of p_2, returning to L before reaching p_2 would require an $H \to L$ crossing, which is impossible.

On (p_2, m) we have $F' > 0$, so Table $(*)$ allows only $1 \to 2$ crossings. Thus product 2 can begin at most once before m. For $x > m$, we have $F' < 0$, so Table $(*)$ allows only $2 \to 1$ crossings, and the product-2 region can end at most once. Hence product 2 is optimal on at most one interval. Endpoint ties at p_2 are treated separately below.



Endpoint ties and threshold choices. It remains to consider ties between the two destinations that occur exactly at p_1 or p_2. These points require a separate argument because one of the drifts vanishes there. The cases in which p_1 or p_2 coincides with a valley or a peak of F have already been covered: a valley at p_1 strictly favors the high destination, a valley at p_2 strictly favors the low destination, and peaks at either endpoint are included in the preceding cases. Suppose first that

$$L(p_1) = H(p_1), \qquad h > p_1.$$

Recall that we have

$$L(p_1) = \frac{F(p_1)}{\delta} \quad \Rightarrow \quad H(p_1) = \frac{F(p_1)}{\delta} \tag{20}$$

Using the bellman equation of the high branch evaluated at p_1 gives

$$\delta H(p_1) = F(p_1) + b_2(p_1)H'(p_1) \quad \Rightarrow \quad b_2(p_1)H'(p_1) = 0$$

And we obtain

$$b_2(p_1) = \lambda D(p_1)(p_2 - p_1) > 0 \quad \Rightarrow \quad H'(p_1) = 0$$

For the low branch, the equation $\delta L = F + b_1 L'$ does not directly determine $L'(p_1)$ because $b_1(p_1) = 0$. We study then the derivative of the equation. Differentiating gives

$$\delta L' = F' + b_1' L' + b_1 L''.$$

using that $b_1(p_1) = 0$ and

$$b_1'(p_1) = -\lambda D(p_1).$$

We obtain,

$$\big(\delta + \lambda D(p_1)\big) L'(p_1) = F'(p_1).$$

Note that a tie is only possible when $F'(p_1) < 0$. Indeed, if $F'(p_1) > 0$, moving to the right toward h yields strictly larger subsequent values of F than holding at p_1, so $H(p_1) > L(p_1)$. On the other hand, if $F'(p_1) = 0$, since $h > p_1$, p_1 is the valley a, and the high destination again strictly improves on holding. Then, we must have $F'(p_1) < 0$. Therefore,

$$L'(p_1) < 0.$$

Hence

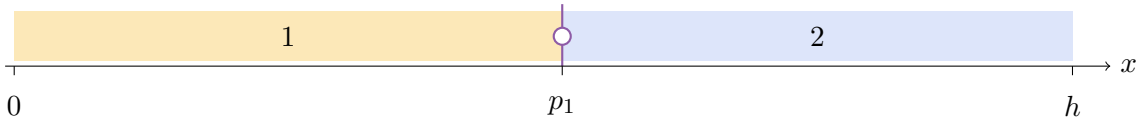
$$(H - L)'(p_1) > 0,$$

so, starting from the tie at p_1, the high branch wins immediately to the right. Thus product 2 is optimal on that side.

It remains to determine the action immediately to the left. By eq (20) we have that $g(p_1) = 0$. Suppose, by contradiction, that product 2 were optimal immediately to the left of p_1. Since $0 < b_1 < b_2$ there, this would imply $W' > 0$ and hence $g = b_2 W' > 0$. On this branch,

$$g' = \frac{\delta g}{b_2} - F' > 0,$$

because $F' < 0$. Thus g would be strictly increasing as x approaches p_1 from the left, which is incompatible with $g > 0$ and $g(p_1) = 0$. Therefore product 1 is optimal immediately to the left of p_1. The tie at p_1 consequently gives a $1 \to 2$ switch.



Now suppose that

$$L(p_2) = H(p_2), \qquad m > p_2.$$

Such a tie requires $F'(p_2) > 0$. At the tie,

$$L'(p_2) = 0, \qquad H'(p_2) > 0,$$

so $(H - L)'(p_2) > 0$. Since we approach p_2 from the left, $H - L < 0$ immediately before the tie, and therefore the low branch wins immediately to the left of p_2.

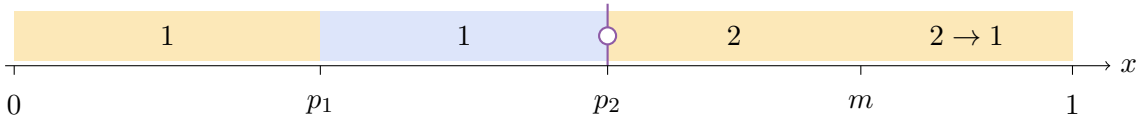
Again,

$$g(p_2) = 0.$$

Suppose, by contradiction, that product 1 were optimal immediately to the right of p_2. Since $b_1 < b_2 < 0$ there, this would imply $W' < 0$ and hence $g = b_1 W' > 0$. On the product-1 branch,

$$g' = \frac{\delta g}{b_1} - F' < 0,$$

because $b_1 < 0$ and $F' > 0$. Hence a positive g cannot emerge to the right from $g(p_2) = 0$. Product 2 is therefore optimal immediately to the right of p_2, and the tie again gives a $1 \to 2$ switch.



At an interior tie between L and H, either destination policy attains the same value, so either may be selected at the tie. If an attractive peak $m \in (p_1, p_2)$ is chosen as the destination, the state can be held at m using the product-2 fraction

$$\frac{m - p_1}{p_2 - p_1}.$$

With these choices at ties and holding thresholds, the optimal policy uses product 2 on at most one interval, completing the proof. □

Proof of Proposition 1. The transformation (6) preserves the ranking of every admissible policy from a given initial state. Lemma 1 and Lemma 2 establish the required payoff shape for the exact Beta demand. The preceding control lemma gives an optimal policy with at most one product-2 interval. Translating back to V leaves that policy unchanged and proves the proposition. □

4. When at least one threshold must exist

An upper bound on the number of thresholds does not guarantee that any threshold exists. The following fixed-policy tests distinguish the two constant-policy cases. Write U_ρ^i for the fluid value obtained by always using product i:

$$U_\rho^i(x) = (R - c_i) \int_0^\infty e^{-\delta t} D(x_i(t; x))\, dt.$$

These functions are continuously differentiable. Indeed, fixed-policy trajectories converge uniformly exponentially to p_i, and their initial state derivatives are positive and eventually decay exponentially. One can therefore differentiate this integral under the integral sign.

Corollary 1 (Existence of a nonconstant threshold policy)**.** *Under the assumptions of Proposition 1, a nonconstant optimal policy is necessary exactly when*

$$\max_{x\in[0,1]} \{-\Delta c + \lambda \Delta p (U_\rho^1)'(x)\} > 0,$$
$$\min_{x\in[0,1]} \{-\Delta c + \lambda \Delta p (U_\rho^2)'(x)\} < 0.$$

When both inequalities hold, there is an optimal policy with either one or two interior thresholds, with one of the three patterns in the proposition.

Proof. The fixed-policy equations are

$$\delta U_\rho^i = D(x)\{R - c_i + \lambda(p_i - x)(U_\rho^i)'\}.$$

If the first maximum is nonpositive, the equation for U_ρ^1 also satisfies the optimal Hamiltonian inequality for product 2. Integrating that inequality along any admissible trajectory proves that its payoff is at most U_ρ^1. Thus always using product 1 is optimal everywhere. Conversely, at a state where the expression is positive, use product 2 for a short time h and then product 1 forever. Its value is

$$U_\rho^1(x) + hD(x)\{-\Delta c + \lambda \Delta p (U_\rho^1)'(x)\} + o(h),$$

which is strictly larger for small $h > 0$. Thus the first strict inequality is equivalent to excluding the globally optimal constant policy 1. The same argument, reversing the roles of the products, shows that the second strict inequality excludes the globally optimal constant policy 2. The proposition then leaves only the three nonconstant patterns. □

5. Interpreting the thresholds

The arrows above describe actions as the state x increases, not a chronological sequence of purchases. At an attractive upper threshold $x_H \in (p_1, p_2)$, holding the state requires the product-2 fraction

$$\frac{x_H - p_1}{p_2 - p_1}.$$

With only pure actions in continuous time, rapid alternation approximates this relaxed value; a pure stationary action cannot hold at such a point. Thus exact attainment must be stated with this convention.

A sustainable interior upper threshold is a local maximum of F and therefore satisfies

$$\left[R - c_1 + \frac{\Delta c}{\Delta p}(p_1 - x_H)\right] D'(x_H) = \frac{\Delta c}{\Delta p}\left[D(x_H) + \frac{\delta}{\lambda}\right].$$

A lower threshold need not satisfy this equation. When it separates a trajectory toward p_1 from one toward a higher destination, the two continuation values coincide but their derivatives can differ. In that case V has a corner, and imposing $V'(x) = \Delta c/(\lambda \Delta p)$ at the threshold would be incorrect. Thresholds outside $[p_1, p_2]$ are passage points rather than sustainable destinations, and are determined by continuation-value matching.

The result proves the complete bound on policy patterns for this fluid model. It does not give elementary closed-form threshold locations or parameter boundaries, and it does not prove the same bound for the original stochastic model.